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4. An alloy is a material composed of a mixture of elements, at least one of which is a metal. The table lists the composition and common uses of different alloys containing silver and gold.

Name of alloy	Composition of alloy by mass (%)	Uses of alloy
amalgam	mercury 48 %, silver 25 %, tin 15 %, copper 12 %	dental fillings, mining
green gold	gold 75 %, silver 6-24 %, copper .... %	Nobel Prize medals, decoration
nordic gold	gold 89 %, aluminium 5 %, zinc 5 %, tin 1 %	coins, decoration
solder	tin 90 %, silver 5 %, copper 5 %	joining electrical components
sterling silver	silver 92.5 %, platinum 4 %, germanium 1.5-3 %, zinc 0.5-2 %	decoration, plumbing, instruments, jewellery
white gold	gold 75 %, palladium 10 %, nickel 10 %, zinc 5 %	decoration, jewellery

(a) **Circle** the number of alloys that contain an element from Group 4 of the Periodic Table. [1]

0 **1** **2** **3** **4** **5** **6**

(b) **Circle** the correct percentage range for the mass of copper that can be found in green gold. [1]

0-25 % **1-19 %** **19-25 %** **6-24 %**

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(c) Tick (✓) the statement that **best** describes the composition of the alloys listed. [1]

- the alloys all contain at least one metal ☐
- the alloys all contain at least two metals ☐
- the alloys all contain at least three metals ☐
- the alloys all contain at least four metals ☐

(d) Tick (✓) the statement that **best** describes the decorative uses of the alloys listed. [1]

- all of the alloys are used for decorative purposes ☐
- all of the alloys containing gold are used for decorative purposes ☐
- all of the alloys containing silver are used for decorative purposes ☐
- none of the alloys containing silver are used for decorative purposes ☐

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(e) A solder joint in an electrical circuit contains 0.00011 kg of silver.

Use this information and the composition of solder given in the table to calculate the mass of tin in the solder joint. [2]

Mass = ... kg

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